



# Environmental Hyperthermia

## Assessment

### Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined by HandTevy.
- Pediatric hypotension is defined as SBP < 70 + (age in years x 2) mmHg

### Signs & Symptoms:

- Weakness
- Nausea & vomiting
- Cramping
- Syncope
- Diaphoresis & anhidrosis
- Altered Mental Status
- Bizarre behavior
- Hypotension
- Tachycardia

### Differential:

- CVA
- Dehydration
- Encephalopathy
- Meningitis / Sepsis
- Head Trauma
- Overdose / Toxin
- Hypoglycemia
- [Hyperactive Delirium with Severe Agitation](#)
- Alcohol withdrawal

## Clinical Management Options

|   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---|---|---|---|---|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P | P | P | P | P | P | <ul style="list-style-type: none"> <li>• Age appropriate core body temperature assessment</li> <li>• <a href="#">Oxygen</a></li> <li>• Move to shaded/cool environment, discontinue physical activity, PO fluids if tolerated</li> <li>• If AMS, then <a href="#">BGL assessment</a></li> <li>• If AMS and/or <b>body temperature &gt; 102.2 F</b>, then active cooling measures per patient condition: <ul style="list-style-type: none"> <li>◦ Ice packs to neck, axilla and groin, wet patient, ice sheets in tactical or rescue environment, and increased airflow</li> </ul> </li> <li>• If <b>body temperature ≥ 105.8 F</b>, then cold water immersion as soon as possible <ul style="list-style-type: none"> <li>◦ Immerse patient 1min/degree Fahrenheit you are attempting to cool, discontinue immersion at 102.2F and move to active cooling</li> </ul> </li> </ul> |
| L | L | L | L | L | L | <ul style="list-style-type: none"> <li>• <a href="#">Vascular access</a></li> <li>• Infuse <a href="#">Isotonic Crystalloid</a> fluids titrated to effect</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1 | 2 | 3 | 4 | 5 | 6 | <ul style="list-style-type: none"> <li>• If AMS, may infuse cold <a href="#">Isotonic Crystalloid</a> if available up to 30 mL/kg or max of 2L</li> <li>• If shivering develops, <a href="#">Midazolam</a> for sedation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## Consult Online Medical Control As Needed

### Pearls:

- Refer to drug formulary charts for all medication dosing for adults and pediatric patients.
- Signs of improvement to help titrate to effect include: improved heart rate, decrease in body temperature, resolution of thirst, feeling the need to urinate and/or increased urination, improvement in mental status, improvement in skin conditions, etc.
- If increased temperature, utilize passive cooling by removing excessive clothing or covers.
- Antipyretics should not be used in the setting of environmental heat emergencies.
- Exertional heat stroke should be suspected in anyone with a history of recent exertion and bizarre behavior, seizure, or syncope.
- Any AMS should have blood glucose performed. Severe heat emergencies may lead to liver dysfunction and hypoglycemia.
- If AMS and cold isotonic crystalloid fluids are unavailable, then ≥PL4 may begin isotonic crystalloid boluses.
- Damage caused by heat stroke is determined by how high the temperature got and how long it remained elevated.
- To make ice sheets: soak a sheet in ice water and drape over the patient's body, change and re-soak q 1min

Cold Isotonic Crystalloid Bolus 30mL/Kg to Max of 2L

| Patient Weight (kg) | 40   | 45   | 50   | 55   | 60   | 65   | 70 ≤ |
|---------------------|------|------|------|------|------|------|------|
| Max mL              | 1200 | 1350 | 1500 | 1650 | 1800 | 1950 | 2000 |